

si koi file block id store kr 32bit usko store krta hai
 jo ki koi file uske id se.

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case 2 - Assume our file is entirely in main memory
 and there is at least space for one extra block

B+BR

SETS OPERATIONS →

(1) Union (2) Intersect (3) Minus / Except.

• Union - merge into single set of rows & columns

R			S		
A	B	C	D	E	F
1	2	3	2	3	4
2	3	4	5	6	7
4	5	6	8	9	10

select * from R
 union select * from S;

name A B C
 1 2 3
 2 3 4
 4 5 6
 5 6 7
 8 9 10

A	B	C
1	2	3
2	3	4
4	5	6
5	6	7
8	9	10

R			S	
A	B	C	E	F
1	2	3	1	2
4	5	6	3	7
7	8	9	6	9
8	11	12	8	11

select A, B from R
 union
 select E, F from S

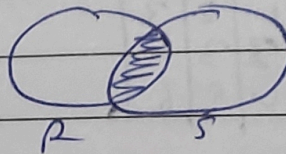
R	
A	B
1	2
4	5
7	8
8	11
3	7
6	9

column
 no. of
 variable
 data

- Imp points about union →
- output print with the making of 1st query
 - By default it is in asc order of first column
 - Number of column in all the queries should be same
 - The data type of the column in each query must be same
 - The name of columns need not be identical in all the queries
 - union can't be used in sub query
 - aggregate fn can't be used with union
 - null values are not ignored
 - In operators has higher precedence than union

union All - for print common content also

Intersect clause -



select * from R
intersect
select * from S

R		S	
A	B	C	D
1	2	1	3
3	4	3	5
5	6	6	8

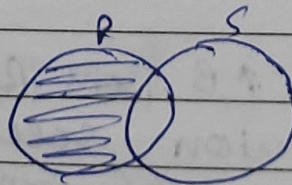
→

S	
A	B
1	2

Imp. pts about Intersect →

- Same as union pts

MINUS CLAUSE →



select * from R minus/except
select * from S

A	B
1	2
5	6

R	
A	B
1	2
3	4
5	6

S	
C	D
3	4
5	7
8	9

Employee
e1Employee
e2

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eno ename manager
noeno ename manager
no

1	KUSH	-
2	LUV	1
3	Raj	1
4	Rohit	2
5	Ram	2

1	KUSH	-
2	LUV	1
3	Raj	1
4	Rohit	2
5	Ram	2

select e1.ename as empname, e2.ename as managername
from Employee as e1, Employee as e2
where e1.m.no = e2.emp.no

emp name	manager name
LUV	KUSH
Raj	KUSH
Rohit	LUV
Ram	LUV

Left JOIN

select e1.ename as empname, e2.ename as managername
from Employee as e1 left join
Employee as e2 on e1.managerno =
e2.emp.no.

JOIN ALGORITHMS →Nested loop
join

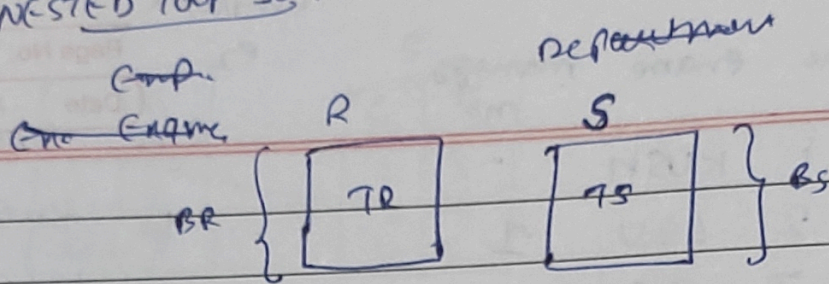
Block nested loop join

Can

NESTED LOOP JOIN

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for each tuple tr in TR do

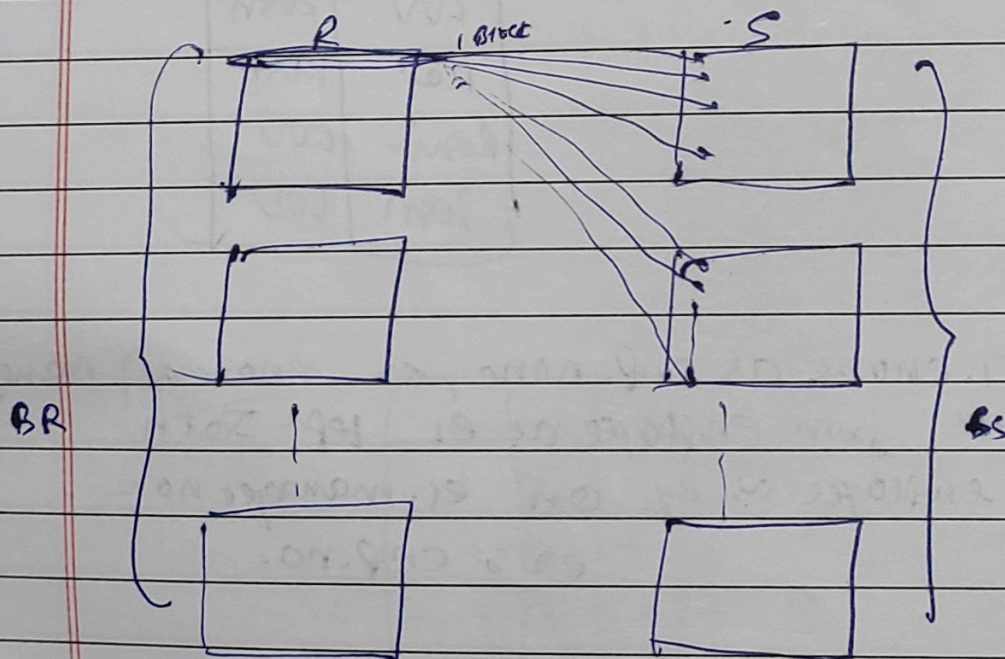
for each tuple ts in TS do

compare (tr, ts) if they satisfies the condition
add them in the result of the join

end

end.

Case 1 - Assume only 2 blocks of main memory available
to store blocks from R & S



TR - total rows in R

Total Blocks transfers needed
 $= TR \times BS + BR$

NO. of blocks = 10
NO. of records = 100

NO. of blocks = 20
NO. of Records = 150

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R is outer & S is inner

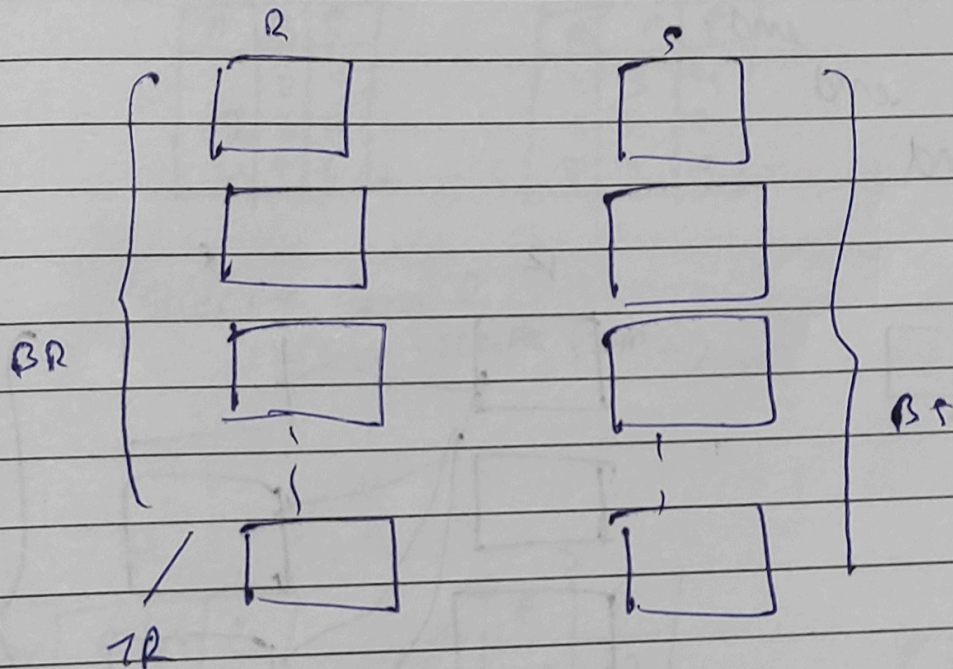
$$100 \times 20 + 10 = 2010 \text{ Ans.}$$

S is outer & R is inner

$$150 \times 10 + 20 = 1520 \text{ Ans.}$$

Rel with less NO. of blocks is inner rel & to minimize total Block transfers

Case 2 - Assume one rel fits entirely in main mem. & there is atleast space for one extra block



$$\text{Total block transfers} = BR + BS$$

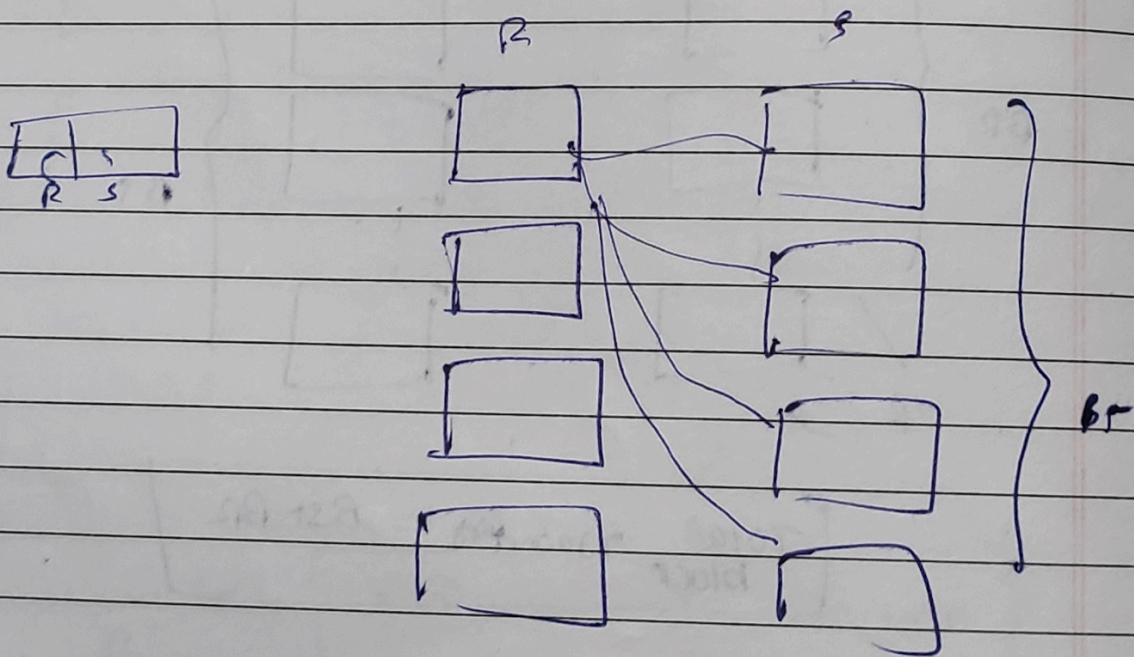
* $\min(BR, BS) + 1$ is the minimum no. of blocks in main memory required to join two relations so that no block is transferred more than once.

In nested loop join, where access cost is req. to join relations if main memory space is very limited.

BLOCK NESTED LOOP JOIN

- For a block of r rows get all tuples in that block and compare with all the tuples of the inner relation.

for each block br in R do
 for each block bs in S do
 for each tuple tr in tr do
 for each tuple ts in ts do
 compare (tr, ts) if they satisfies the condition
 add them in the result of the join
 end
 end
 end
 end



$$[B_R * B_S + B_R]$$

R
= 10 blocks

S
= 20 blocks

S is inner

R is outer, S is inner

$$10 * 20 + 10 \\ = 210$$

$$20 * 10 + 20$$

$$= 220 \checkmark$$